

Data Structures and Algorithms -3 (25CS2103E)

Product Search and Recommendation System for Shopping

BY TEAM 4

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Abstract

The Product Search and Recommendation System for Shopping is designed to provide a smarter and more efficient way for users to search and discover products on an online shopping platform. With the rapid growth of e-commerce, customers often find it difficult to locate the right product due to incorrect spellings, incomplete product names, or a large number of similar products. The proposed system overcomes these challenges by enabling users to search using product names, categories, brands, keywords, or descriptive queries. It retrieves the most relevant products from a structured product repository and presents them in an organized and user-friendly manner, making the search process faster and more accurate.

In addition to efficient product search, the system recommends similar or related products based on their category, brand, features, and user preferences. It also allows users to apply filters such as price range, brand, ratings, and availability to narrow down their search results and make informed purchasing decisions. By combining intelligent search with personalized recommendations, the system enhances the overall shopping experience, reduces the time spent searching for products, and helps users easily discover products that best match their needs. The proposed solution is scalable, reliable, and suitable for modern e-commerce applications that manage large product collections.

Introduction

Online shopping has become an important part of everyday life, with e-commerce platforms offering thousands of products in different categories, brands, and price ranges. As the number of products increases, users may find it difficult and time-consuming to locate the exact product they need. Searching becomes even more difficult when users enter incomplete product names, spelling mistakes, or describe their requirements instead of knowing the exact product name.

The **Product Search and Recommendation System for Shopping** aims to provide a simple and intelligent solution to these problems. The system allows users to search for products using keywords or descriptive queries and provides features such as spelling correction, auto-complete, and product filtering. It also supports intent-based searches such as *“laptop for gaming under ₹80,000”* by identifying the user's basic requirements and finding suitable products. In addition, the system recommends relevant products and provides reasons for the recommendations, helping users compare options and make better purchasing decisions.

Literature Survey

Existing e-commerce platforms provide product search and recommendation features to help users find products from large collections. Traditional search systems mainly match user-entered keywords with product names, descriptions, and categories. Modern platforms also provide features such as **filters, auto-complete, spelling correction, and product recommendations** to improve the shopping experience. These features help users narrow down large product collections and find suitable products more easily. However, users may still receive irrelevant results when they enter spelling mistakes, incomplete queries, or describe their requirements instead of using an exact product name.

The proposed system builds upon these existing approaches by combining **smart product search and explainable recommendations** in a single system. It focuses on handling spelling errors, providing search suggestions, applying filters based on price, brand, and category, and understanding requirement-based queries such as *“gaming laptop under ₹80,000.”* The system identifies important details from the user's query, searches the available product collection, and ranks products according to how closely they match the requirements. It also explains why each product is recommended, helping users understand the results, compare suitable options, and make better purchasing decisions.

Problem Statement and Objectives

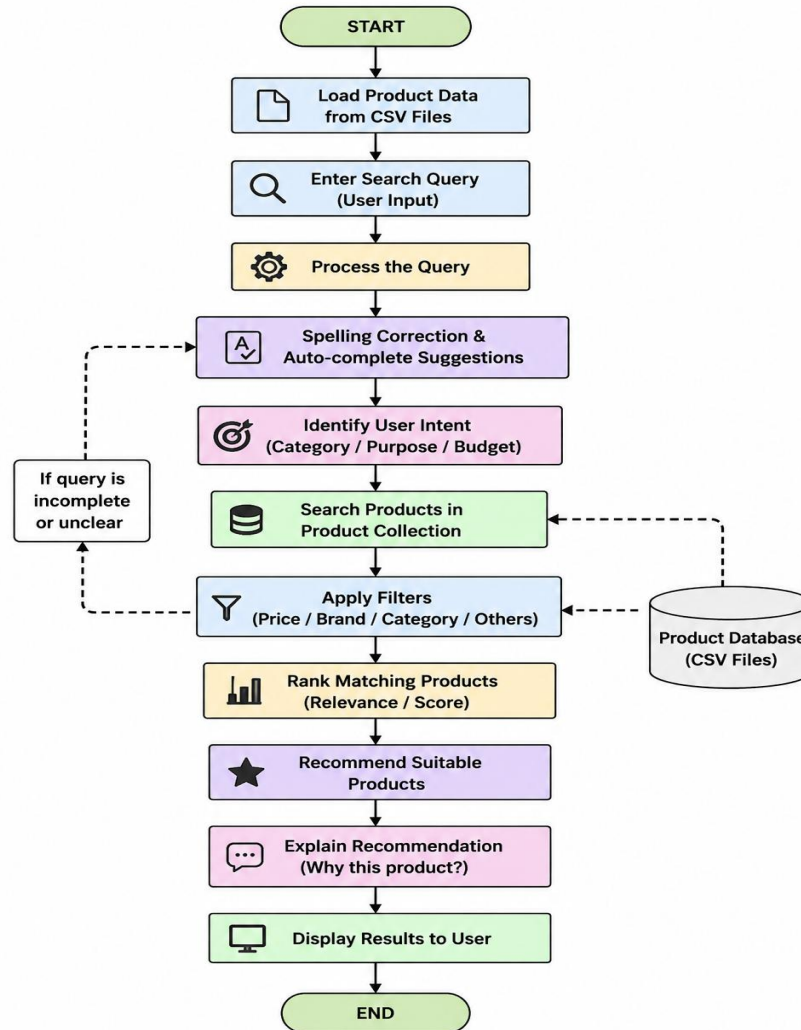
Problem Statement

Modern e-commerce platforms contain millions of products, making it difficult for customers to quickly find the items they need. Traditional search systems often fail when users enter incomplete product names, spelling mistakes, or search using natural language such as "laptop for programming under ₹60,000." As a result, users receive irrelevant search results, spend more time searching, and may fail to discover suitable alternatives. There is a need for an intelligent product search and recommendation system that understands user intent, retrieves relevant products efficiently, and recommends similar products to improve the overall shopping experience.

Objectives

- To provide fast and efficient product searching.
- To handle common spelling mistakes in user queries.
- To provide auto-complete suggestions while searching.
- To filter products based on price, brand, and category.
- To understand simple user requirements from descriptive search queries.
- To identify and rank products that best match the user's requirements.
- To recommend suitable and related products.
- To explain why each product is recommended.
- To improve the overall product discovery and shopping experience.

Flow Chart



Results

The system provides fast and relevant product search using **price, brand, category, and user requirements**. It handles spelling mistakes and suggests suitable products. Results include **product details, ratings, match scores, and recommendations**, helping users compare and select products easily.

```
PS C:\Users\student\SmartShopping> java -cp bin Main
=====
PRODUCT SEARCH AND RECOMMENDATION SYSTEM FOR SHOPPING
=====
Loading catalog ... 2902 products loaded.

1. Login
2. Register
3. Exit
Choose: 1
Username: testuser
Password: pass123
Login successful. Session saved to data/session.txt
=====

SELECT A CATEGORY
=====
1. Electronics      2. Home Appliances
3. Beauty           4. Fashion
5. Grocery           6. Furniture
7. Sports & Fitness  8. Books & Stationery
9. Automotive       10. Toys & Games
11. Kitchen         12. Footwear
13. Bags & Accessories 14. Jewelry & Watches
15. All Products
0. Logout and Exit
=====
Choose category: 1
=====
PRODUCTS IN ELECTRONICS
=====
ID   NAME                                     TYPE   PRICE
1001 Samsung Smartphone Prime 100         Smartphone Rs 10149
1002 Xiaomi Laptop Pro 107                 Laptop    Rs 8959
1003 HP Headphones Edge 114                 Headphones Rs 10745
1004 Realme Smartwatch Elite 121             Smartwatch Rs 19854
1005 OnePlus Tablet Elite 128               Tablet    Rs 13049
1006 Dell Power Bank Pro 135                Power Bank Rs 5010
1007 Boat Bluetooth Speaker Nova 142        Bluetooth Speak. Rs 8680
1008 Apple Monitor Neo 149                  Monitor   Rs 4599
1009 Sony Keyboard Neo 156                  Keyboard  Rs 4576
1010 Lenovo Mouse Pro 163                   Mouse     Rs 550
... showing 10 products. Use search to narrow down.
=====
CATEGORY: Electronics | Cart: 0 item(s)
=====
1. Search by requirement / description (main feature)
2. Select a product by ID (view details & add to cart)
3. Apply Filters
4. Recommendations
5. View Cart
6. Show more products in this category
7. Back to Categories
8. Exit
=====
Choose:
```

```
PS C:\Users\student\SmartShopping> java -cp bin Main
Describe what you need. Example:
"lightweight laptop with long battery life for students under 45000"
Your requirement: i want a budget friendly smartphone
=====
MATCHING PRODUCTS (10 shown, searched in 77 ms)
Requirement keywords : smartphone
Detected budget      : none
=====
```

ID	NAME	TYPE	PRICE	SCORE	SUITABLE
1001	Samsung Smartphone Prime 100	Smartphone	Rs 10149	137	YES
1085	Lenovo Smartphone Pro 688	Smartphone	Rs 8889	127	YES
1061	Dell Smartphone Zen 520	Smartphone	Rs 20859	127	YES
1037	Xiaomi Smartphone Air 352	Smartphone	Rs 15819	120	YES
1049	Sony Smartphone Neo 436	Smartphone	Rs 30939	105	YES
1097	Boat Smartphone Core 772	Smartphone	Rs 32199	104	YES
1109	Realme Smartphone Core 856	Smartphone	Rs 32829	103	YES
1073	HP Smartphone Prime 604	Smartphone	Rs 37239	99	YES
1013	Apple Smartphone Lite 184	Smartphone	Rs 42909	93	YES
1025	OnePlus Smartphone Pro 268	Smartphone	Rs 46689	90	YES

Enter product ID for full details (0 to go back):

```
PS C:\Users\student\SmartShopping> java -cp bin Main
Enter product ID for full details (0 to go back): 1085
=====
COMPLETE PRODUCT DETAILS
=====
Product ID : 1085
Name       : Lenovo Smartphone Pro 688
Brand      : Lenovo
Category   : Electronics
Type       : Smartphone
Price      : Rs 8889
Description:
This smartphone from Lenovo is a professional grade performance
designed for daily use. It offers lightweight and portable, great
for online classes and durable metal body. Ideal choice for
customers looking for a reliable smartphone in the electronics
category at around 8889 rupees.

===== WHY THIS PRODUCT? =====
Suitability : YES (relevance score 127)
=====
1) MATCHING KEYWORDS / FEATURES
   smartphone
   (found in the product description)
   covered 1 of 1 requirement keywords
2) DESCRIPTION MATCH
   longest common text segment with your requirement : 11 characters
   local alignment score : 23
3) CATEGORY / TYPE MATCH
   category : Electronics [matches your selection/query]
   type      : Smartphone [you asked for this type]
4) BUDGET / PRICE MATCH
   price Rs 8889 (no budget was specified in your requirement)
5) REQUIREMENT MATCH - FINAL DECISION
   YES - this product satisfies most of what you asked for, and the price fits.
=====

1. Add to Cart
2. Back to results
3. Leave (exit application)
Choose:
```



```
PS C:\Users\student\SmartShopping> java -cp bin Main
Describe what you need. Example:
    "lightweight laptop with long battery life for students under 45000"
Your requirement: smartfone with excelent camra under 30000
Did you mean: "smartphone with excellent camera under 30000" ?
Use the corrected query? (y/n): y

-----
MATCHING PRODUCTS  (4 shown, searched in 64 ms)
Requirement keywords : smartphone excellent camera 30000
Detected budget      : Rs 30000

-----

ID   NAME                                TYPE    PRICE  SCORE  SUITABLE
1061 Dell Smartphone Zen 520             Smartphone Rs 20859 95    PARTIALLY
1037 Xiaomi Smartphone Air 352           Smartphone Rs 15819 94    PARTIALLY
1001 Samsung Smartphone Prime 100        Smartphone Rs 10149 92    PARTIALLY
1085 Lenovo Smartphone Pro 688            Smartphone Rs 8889 91    PARTIALLY

-----

Enter product ID for full details (0 to go back):
```

```
PS C:\Users\student\SmartShopping> java -cp bin Main
-- RECOMMENDATIONS --
1. Best bundle inside a budget
2. Match my requirement aspects
3. Surprise picks from this category
4. Similar to a product I liked
5. Back
Choose: 2
Enter your requirement aspects separated by commas.
Example: long battery, lightweight, good camera
Aspects: long battery, lightweight, gaming

-----
ASPECT -> PRODUCT ASSIGNMENT
-----
long battery      -> Dell Camera Zen 590           Rs 78198
lightweight       -> Samsung Smartphone Prime 100    Rs 10149
gaming            -> Apple Laptop Prime 779          Rs 66919
Maximum matching size = 3 of 3 aspects.

=====
CATEGORY: Electronics | Cart: 0 item(s)
-----
1. Search by requirement / description  (main feature)
2. Select a product by ID (view details & add to cart)
3. Apply Filters
4. Recommendations
5. View Cart
6. Show more products in this category
7. Back to Categories
8. Exit

-----
Choose:
```

```
PS C:\Users\student\SmartShopping> java -cp bin Main
-- RECOMMENDATIONS --
1. Best bundle inside a budget
2. Match my requirement aspects
3. Surprise picks from this category
4. Similar to a product I liked
5. Back
Choose: 3

-----
SURPRISE PICKS
-----
1077 Boat Tablet Edge 632                Rs 55169 Tablet
1086 Samsung Laptop Zen 695              Rs 25339 Laptop
1122 Xiaomi Laptop Plus 947              Rs 10219 Laptop
1083 HP Camera Neo 674                   Rs 90438 Camera
1027 Boat Headphones Edge 282             Rs 17045 Headphones

=====
CATEGORY: Electronics | Cart: 0 item(s)
-----
1. Search by requirement / description  (main feature)
2. Select a product by ID (view details & add to cart)
3. Apply Filters
4. Recommendations
5. View Cart
6. Show more products in this category
7. Back to Categories
8. Exit

-----
Choose:
```



```
PS C:\Users\student\SmartShopping> java -cp bin Main
```

```
-- RECOMMENDATIONS --
```

1. Best bundle inside a budget
2. Match my requirement aspects
3. Surprise picks from this category
4. Similar to a product I liked
5. Back

```
Choose: 4
```

```
Enter a product ID you liked: 1002
```

```
Finding products similar to: Xiaomi Laptop Pro 107
```

```
-----  
SIMILAR PRODUCTS
```

1062	Boat Laptop Sport 527	Rs 21559	similarity 394
1074	Realme Laptop Core 611	Rs 93379	similarity 247
1086	Samsung Laptop Zen 695	Rs 25339	similarity 247
1014	Sony Laptop Plus 191	Rs 90859	similarity 247
1038	HP Laptop Ultra 359	Rs 36679	similarity 247

```
=====
```

```
CATEGORY: Electronics | Cart: 0 item(s)
```

- ```

```
1. Search by requirement / description (main feature)
  2. Select a product by ID (view details & add to cart)
  3. Apply Filters
  4. Recommendations
  5. View Cart
  6. Show more products in this category
  7. Back to Categories
  8. Exit
- ```
-----
```

```
Choose:
```

```
PS C:\Users\student\SmartShopping> java -cp bin Main
```

```
Choose: 1
```

```
Product ID to add: 1004
```

```
Added.
```

```
-----  
YOUR CART (testuser)
```

ID	NAME	QTY	PRICE	SUBTOTAL
1002	Xiaomi Laptop Pro 107	1	Rs 8959	Rs 8959
1003	HP Headphones Edge 114	1	Rs 10745	Rs 10745
1004	Realme Smartwatch Elite 121	1	Rs 19854	Rs 19854
running total			: Rs 39558	[verified]
cart total			: Rs 39558	

```
-----
```

```
TOTAL ITEMS : 3
```

```
TOTAL PRICE : Rs 39558
```

```
shortest store pickup route: Electronics -> Electronics -> Electronics
```

- ```

```
1. Add a product by ID
  2. Delete a product by ID
  3. Checkout (generate order ID)
  4. Continue shopping
  5. Back to categories
  6. Exit

```
Choose: 3
```

```
PS C:\Users\student\SmartShopping> java -cp bin Main
```

```
Choose: 3
```

```
ORDER PLACED
```

```
Order ID : ORD-812431 (prime, verified)
```

```
Items : 3
```

```
Amount : Rs 39558
```

```
Customer : testuser
```

```
Order saved to data/orders.txt. Cart cleared.
```



# Hardware and Software requirements

## Hardware Requirements

- **Processor:** Intel Core i3 or equivalent and above
- **RAM:** Minimum 4 GB
- **Storage:** Minimum 10 GB free space
- **Display:** Standard monitor
- **Input Devices:** Keyboard and mouse

## Software Requirements

- **Programming Language:** Java
- **JDK:** JDK 17
- **IDE:** Visual Studio Code (VS Code)
- **Java Extension:** Extension Pack for Java
- **Data Storage:** CSV / Text Files
- **Operating System:** Windows
- **Version Control:** GitHub

## Conclusion

The **Product Search and Recommendation System for Shopping** demonstrates how a simple product repository can be transformed into a more useful and intelligent shopping assistant. Instead of relying only on exact product names, the system considers what the user is actually trying to find and helps narrow down the available choices.

The project also shows how different algorithmic techniques can be applied to solve practical problems in e-commerce. Features such as search assistance, requirement-based matching, product ranking, and recommendation explanations make the system more useful from a user's perspective. Since the system is implemented using **Java and file handling**, it remains simple to develop and understand while providing a strong foundation for future improvements.

Overall, the project provides a practical combination of **search, decision support, and recommendation**, with the goal of making product discovery quicker, clearer, and more convenient for users.

## Future Scope

- The system can be further developed to provide a more advanced and personalized shopping experience. Future improvements can include:
  - **Voice-based search** for users who prefer speaking instead of typing.
  - **Image-based search** to find products using an uploaded product image.
  - **Personalized recommendations** based on user search and purchase history.
  - **Advanced AI-based intent understanding** for more complex natural-language queries.
  - **Real-time price comparison** across different shopping platforms.
  - **Price-drop notifications** when the price of a selected product decreases.
  - **Online database integration** to handle a much larger and continuously changing product collection.
  - **Mobile application** so users can access the system from smartphones.
  - **User feedback integration** to improve future recommendations.
  - **Multilingual search** to allow users to search for products in different languages.

These enhancements can transform the current system from a basic intelligent search application into a more complete **personalized shopping assistant**.

**Thank you**